

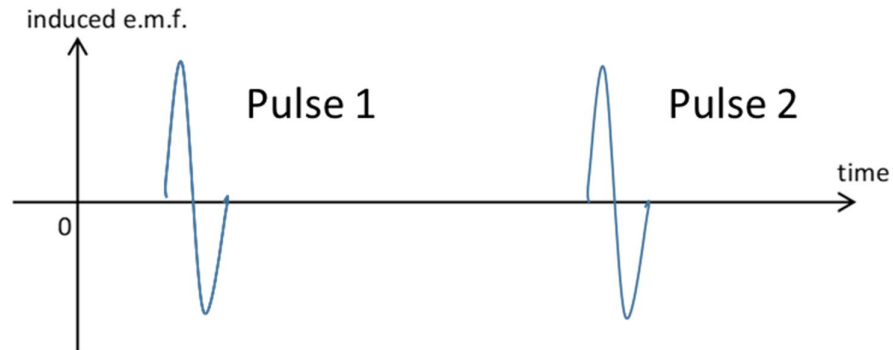
- 5(a) Lenz's law states that the induced emf and hence current in conductor will be such that it opposes the change causing it.

Using a clamp and retort stand, rigidly hold a coil connected to a galvanometer in a closed circuit with a switch. Freely hang a magnet as an oscillating pendulum from another retort stand and allow the magnet to oscillate such that its plane of oscillation brings the magnet closer and further from one end of the coil.

As the magnet approaches, an induced current will flow in the coil, producing a magnetic field that opposes the motion of the magnet, slowing it down, this can be seen as a deflection in the galvanometer. As the magnet oscillates away, the induced current will flow in the coil in the opposite direction to produce a magnetic field that attracts the magnet, further slowing it down.

- (b)(i) As the projectile approaches the coils, the magnetic field experienced at each coil becomes larger, the closer the projectile is. The coils thus experience changing magnetic flux and hence according to Faraday's law, experience an emf.

(ii)



(iii)
$$v = \frac{d}{t} = \frac{1.50}{2.40 \times 10^{-3}} = 625 \text{ m s}^{-1}$$

- (iv) The pulses will be closer together as the bullet takes a shorter time to travel between the coils.

The amplitude of the induced emf of each pulse will be larger as the coils will experience a higher rate of change of magnetic flux linkage.

6(a) Spontaneous means that the decay process occurs on its own accord and